Question:

There exist real numbers x and y, both greater than 1, such that
$\log_x\left(y^x\right)=\log_y\left(x^{4y}\right)=10$. Find xy.

Reasoning + Answer:

<think>

1. Given the equation $\log_x(y^x) = 10$, let's reorganize it using the logarithm property $\log_b
(a^c) = c\log_b(a)$.
Thus, $\log_x(y^x) = x\log_x(y) = 10$.

Similarly, $\log_y(x^{4y}) = 4y\log_y(x) = 10$.

2. Let $a = \log_x(y)$. The given equations can now be written as:
$xa = 10$
$4ya = 10$
From $xa = 10$, we solve for x:
$a = \frac{10}{x}$

Using this value for a in the second equation:
$4y\left(\frac{10}{x}\right) = 10$
Dividing both sides by 10:
$4y\cdot\frac{1}{x} = 1$
This simplifies to:
$\frac{4y}{x} = 1$
Multiplying both sides by x:
$4y = x$
3. Now, substitute $y = \frac{x}{4}$ into $xa = 10$:
Using $a = \frac{10}{x}$:
$x\cdot\frac{10}{x} = 10$

This is consistent with the value of a.

4. From $\frac{x}{4} = \log_x(y)$, and using our value of $y = \frac{x}{4}$:
$\frac{x}{4} = \log_x\left(\frac{x}{4}\right)$

We have:
\[
\left(\frac{x}{4}\right)^5 = x
\]

So solve $\left(\frac{x}{4}\right)^5 = x$:
Let $z = x/4$, then we have:
$z^5 = x$

Substituting, solve:

$z^5 = 4z$
Solving algebraically:

$z^5 - 4z = 0$
$z(z^4 - 4) = 0$

Hence, $z=2^{1/2} or$
Knowing z is a real root, we recheck:

So given fundamental formula (root value), we provide:
$z = \sqrt{2}$ (real value)
verify root:

Now, we have $ xz = 4\sqrt2 = 4\sqrt2
substitution a, b = with order,
x = y

$xy = x^2$
$ = 4\sqrt2
</think>
<answer>
1. $xy = 2\cdot8 = 2^3\cdot2^{1/2}=2^{5}$
3. $xy=64 (final)$
3. Recalculated, box properly 16\end{answer> <answer>16</answer>